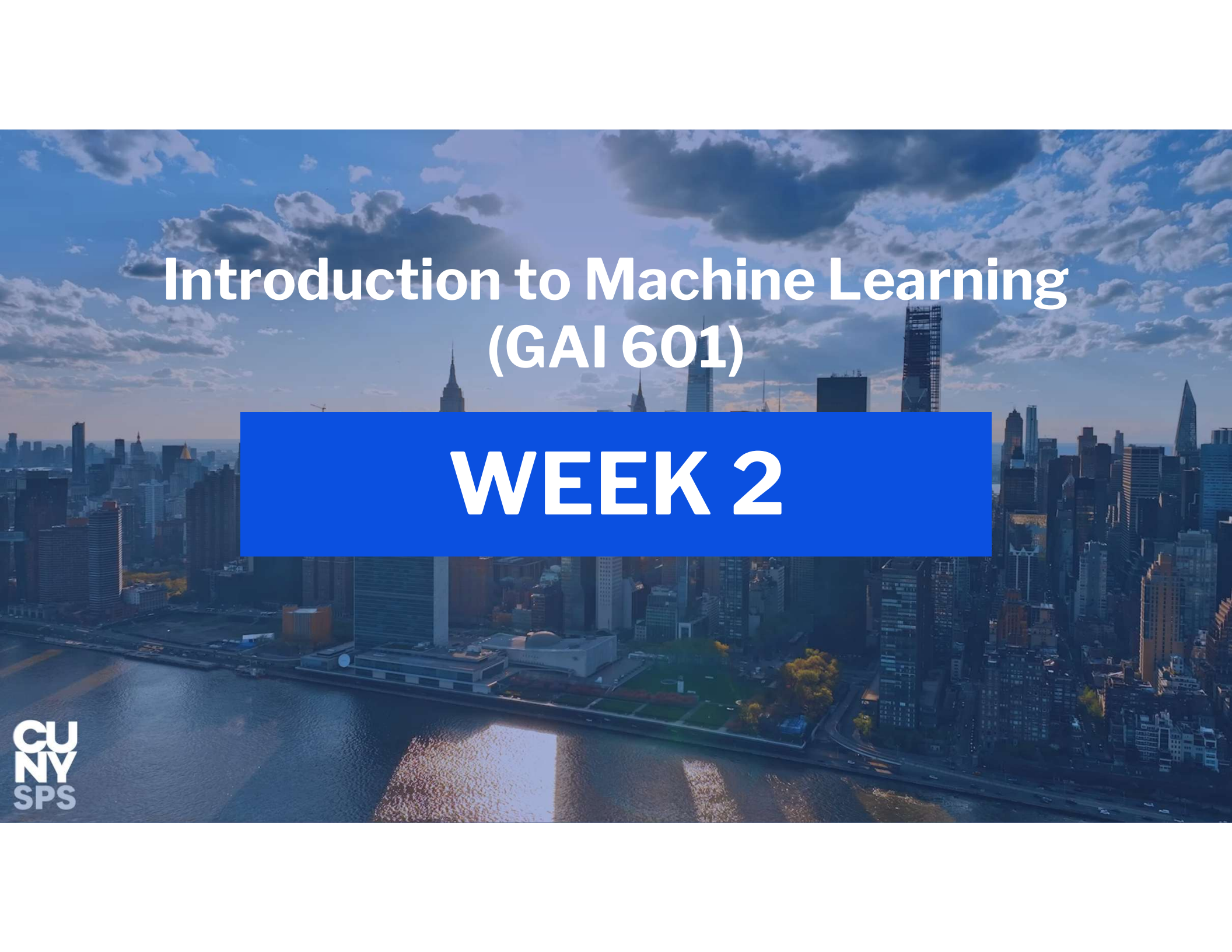
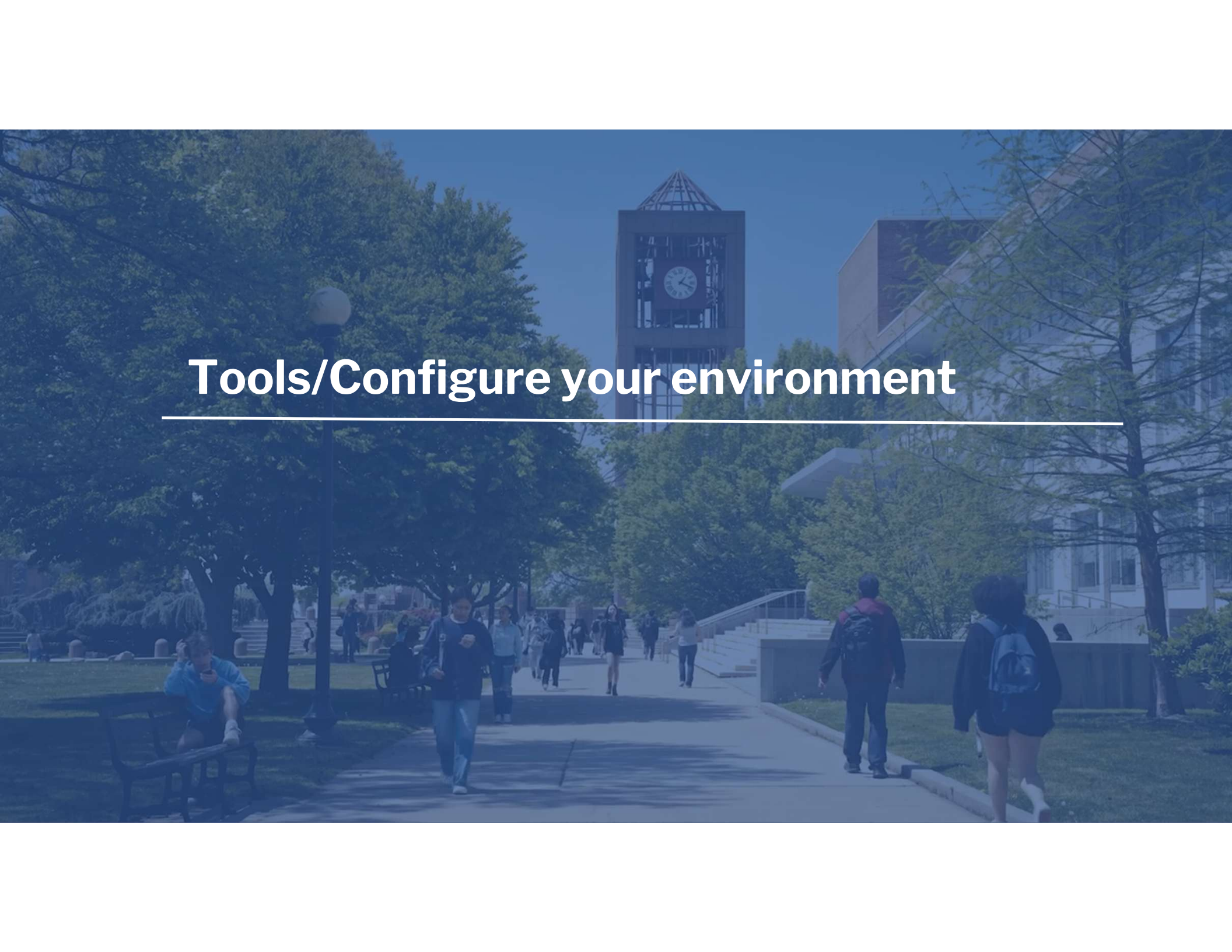


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large blue rectangular box is superimposed over the center of the image, containing the text 'WEEK 2'.

Introduction to Machine Learning (GAI 601)

WEEK 2

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground shows a wide, paved pedestrian walkway lined with mature trees. Several students are walking along the path; some are carrying backpacks. On the left, a student is sitting on a wooden bench. The overall atmosphere is that of a busy academic environment.

Tools/Configure your environment

Register for free tools

1. Colab

- Go to <https://colab.research.google.com/signup>
- Create a Google account (if you don't have one)
- Sign up for **Colab Pro for Education** (free for students)

2. GitHub Copilot

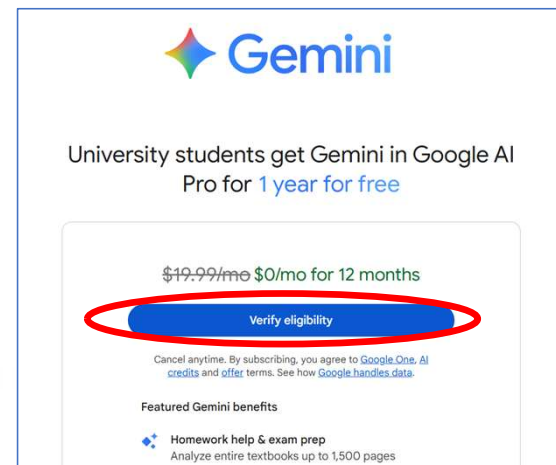
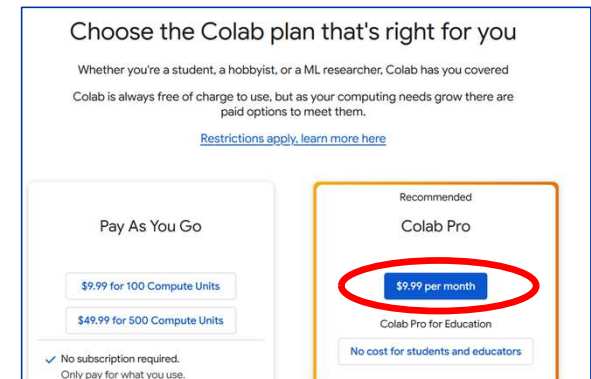
- Go to <https://github.com/education>
- Click on **Join GitHub Education**
- Create a GitHub account (if you don't have one)
- Click on Start an Application to get access

3. Google Gemini

- Go to <https://gemini.google/us/students>
- Fill out the verification form

4. Microsoft Copilot 365

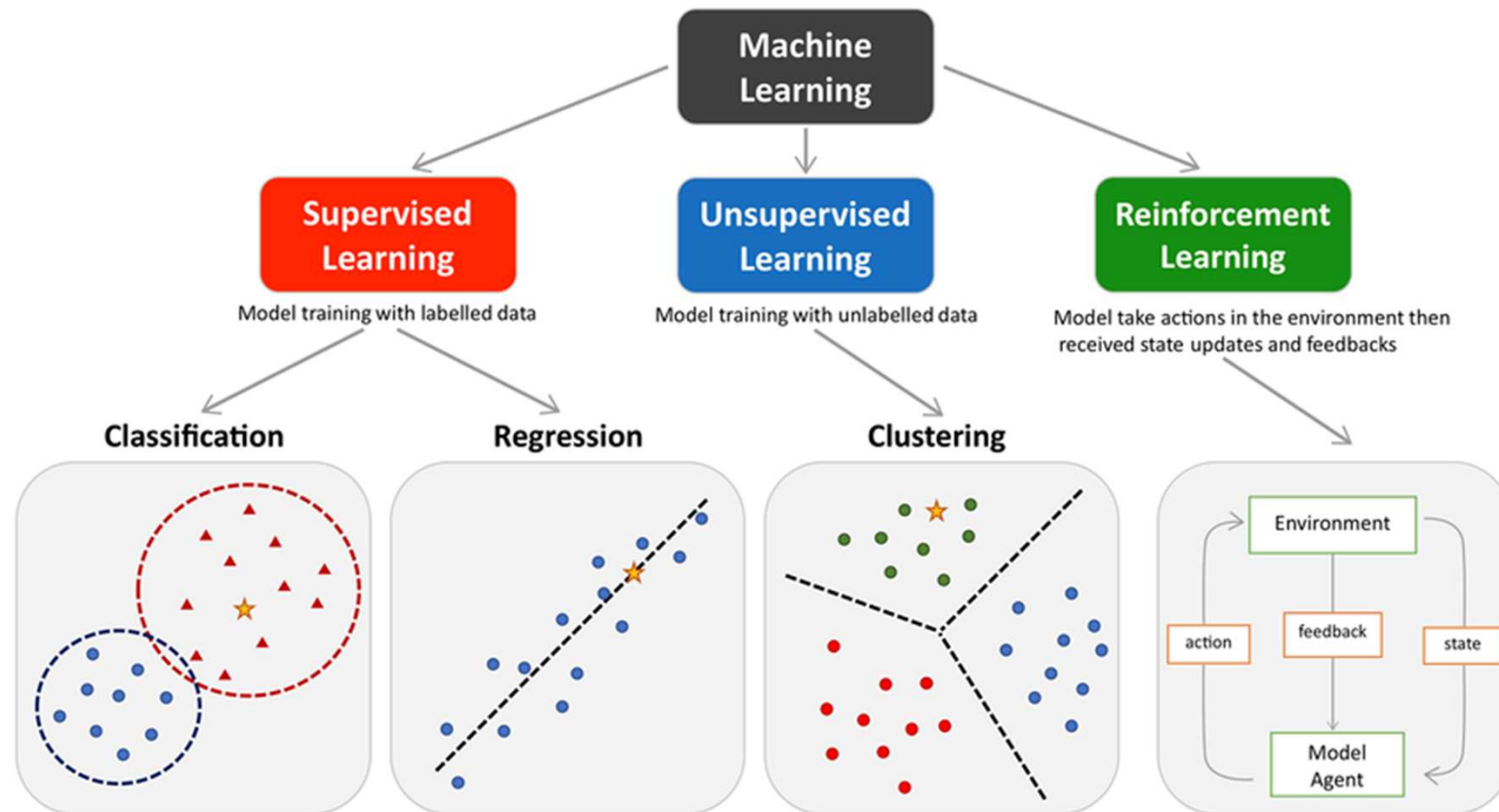
- Free with you Office 365 access (via CUNY)



Machine Learning



Types of Machine Learning

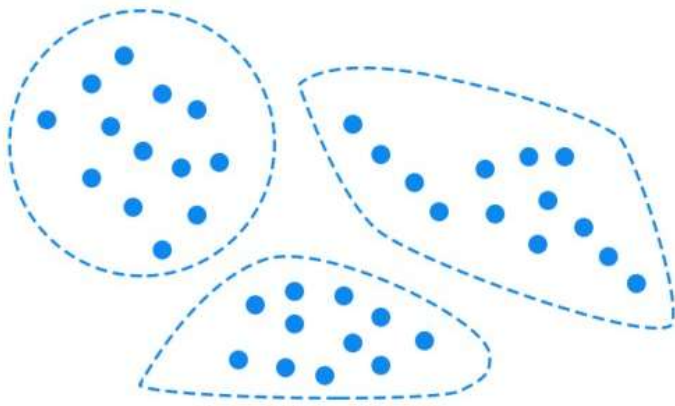


Source: "Machine Learning Techniques for Personalised Medicine Approaches in Immune-Mediated Chronic Inflammatory Diseases: Applications and Challenges", Pend et al, Sep 2021

Supervised vs Unsupervised Learning

UNSUPERVISED

Clustering

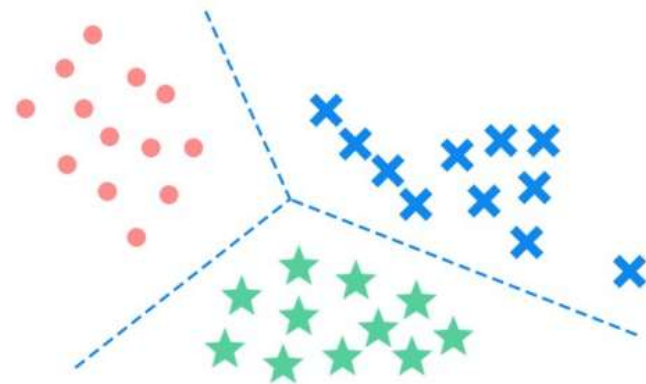


Business use-cases:

- Customer segmentation
- Cross- selling
- Anomaly detection e.g. Expenses
- Document clustering
- Recommendation & personalization
- Data Reporting (dimensionality Reduction)

SUPERVISED

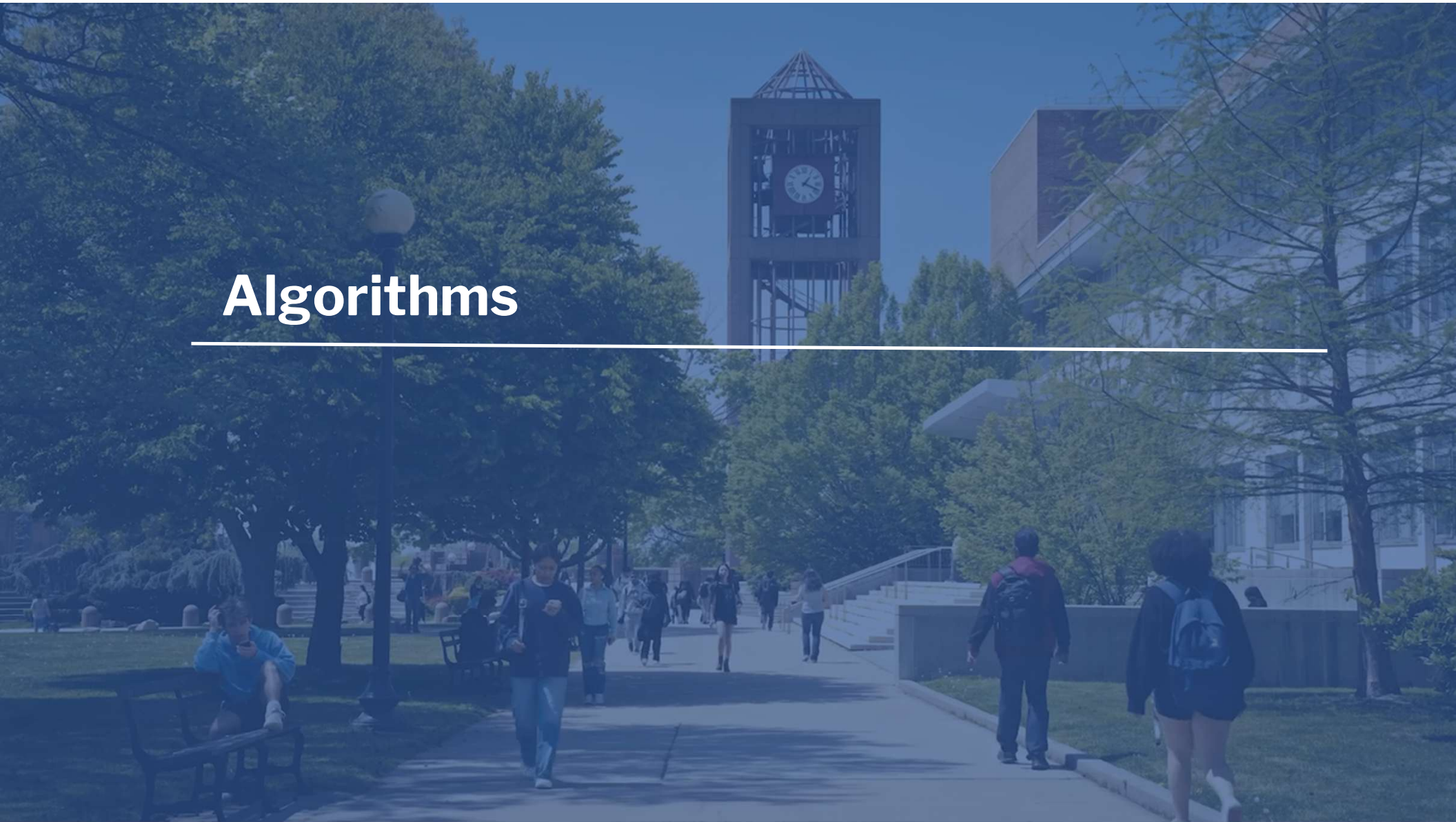
Classification



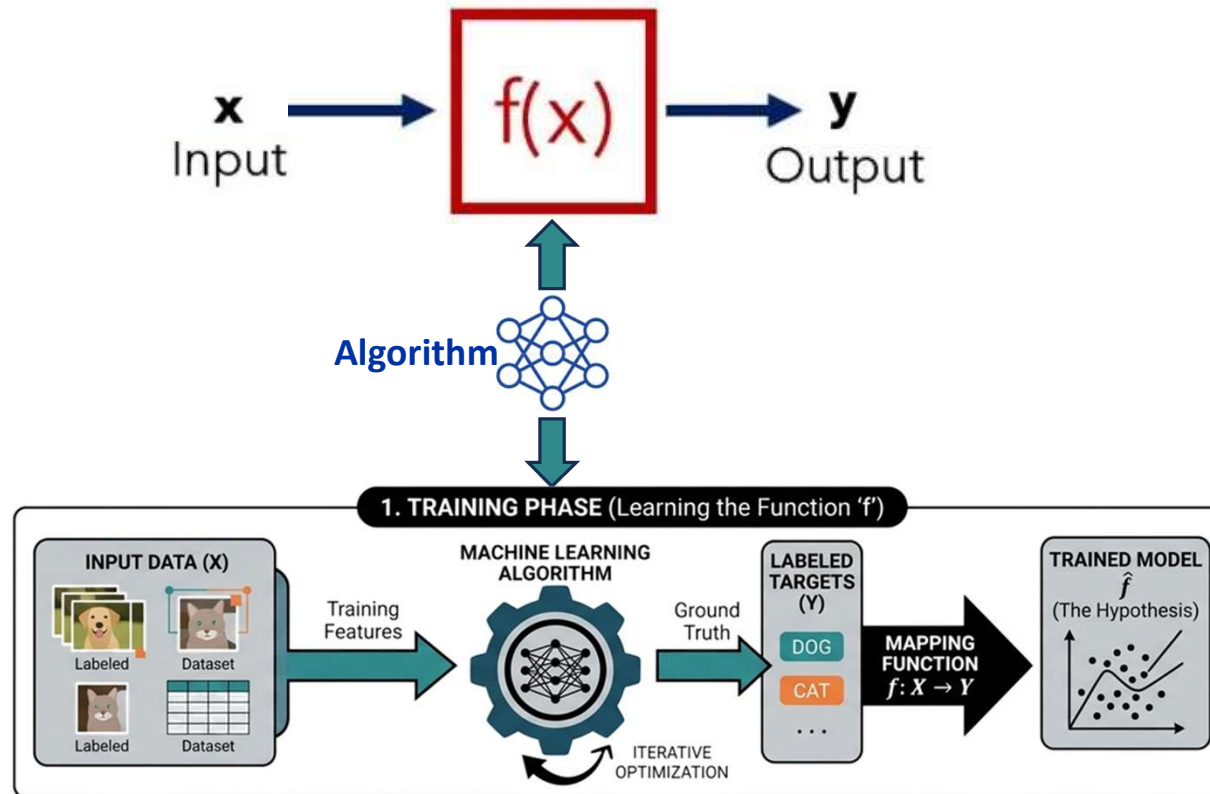
Business use-cases:

- Loan approval, credit risk assessment
- Credit Card Fraud Detection
- Sentiment analysis, text classification
- Speech recognition, Image classification
- Spam Detection
- Stock Price Prediction

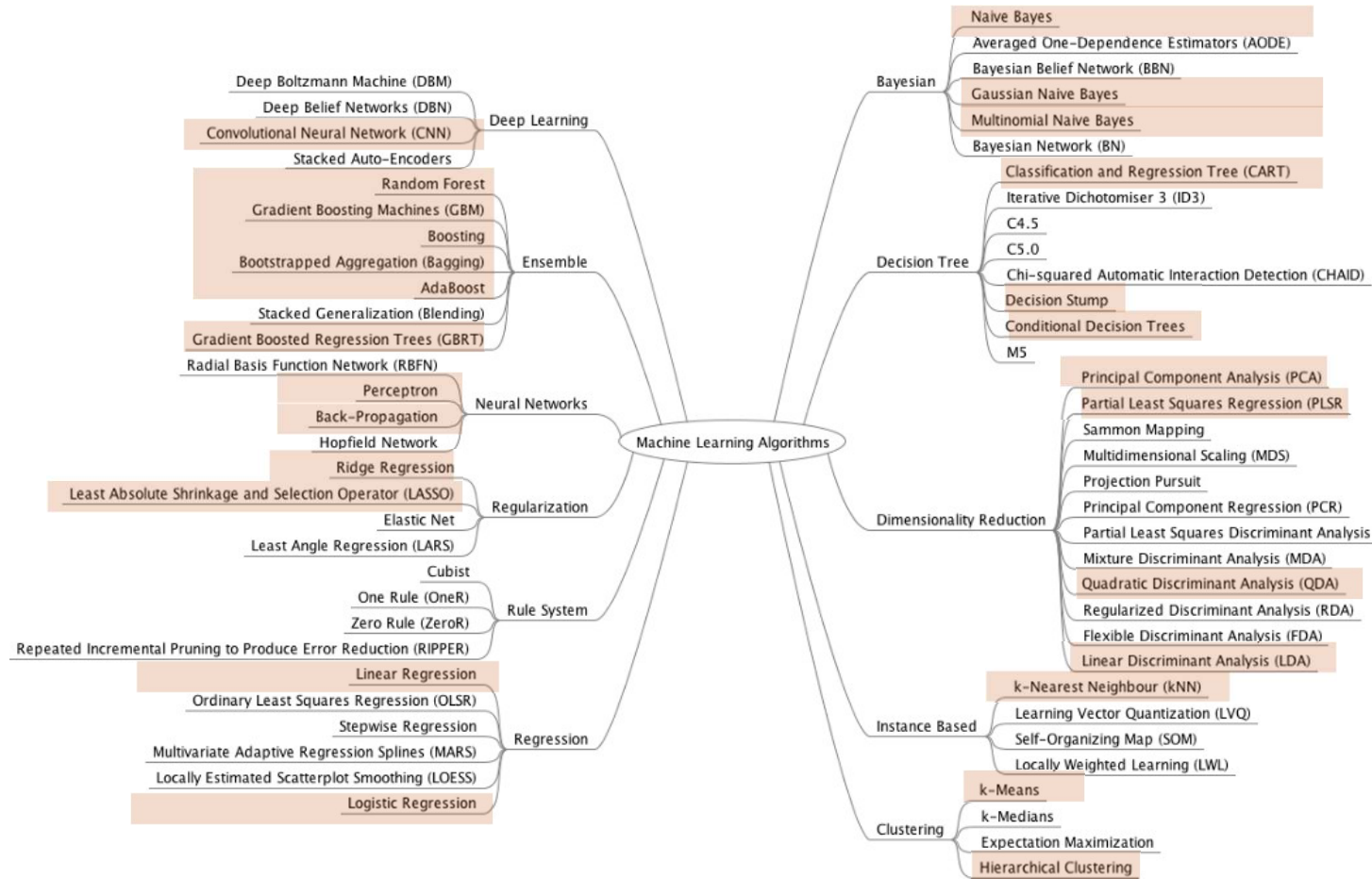
Algorithms



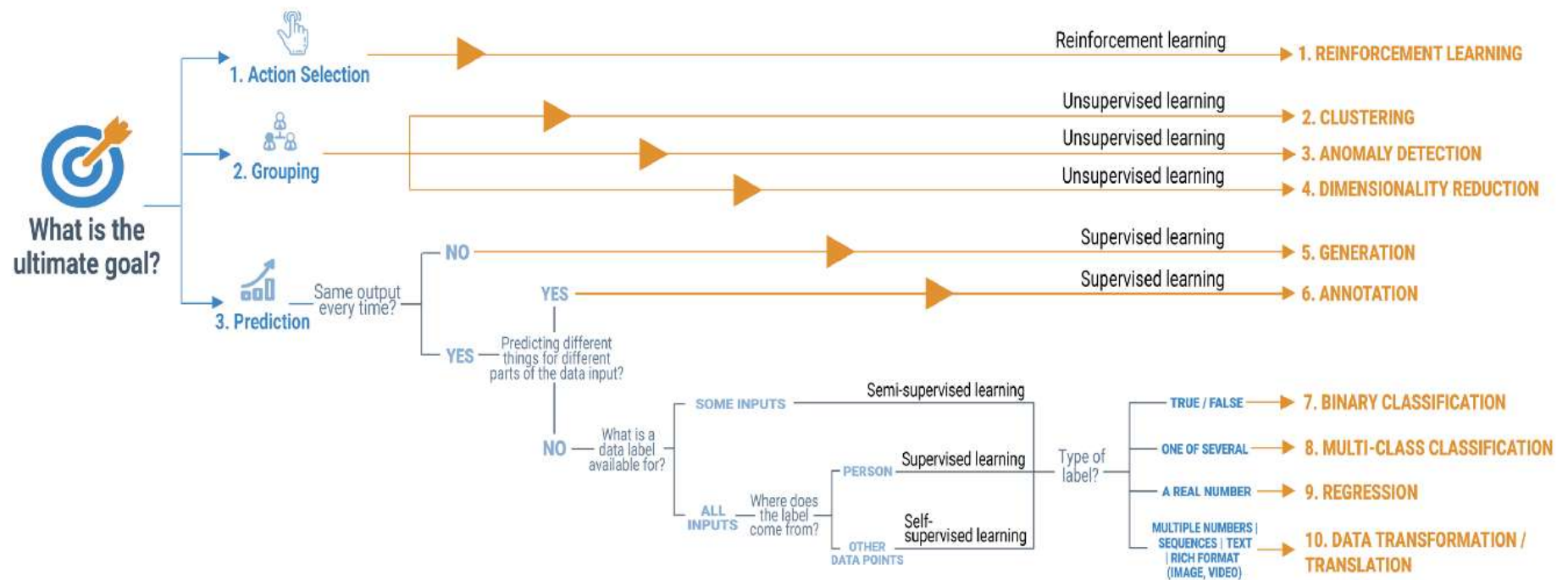
ML as a Function



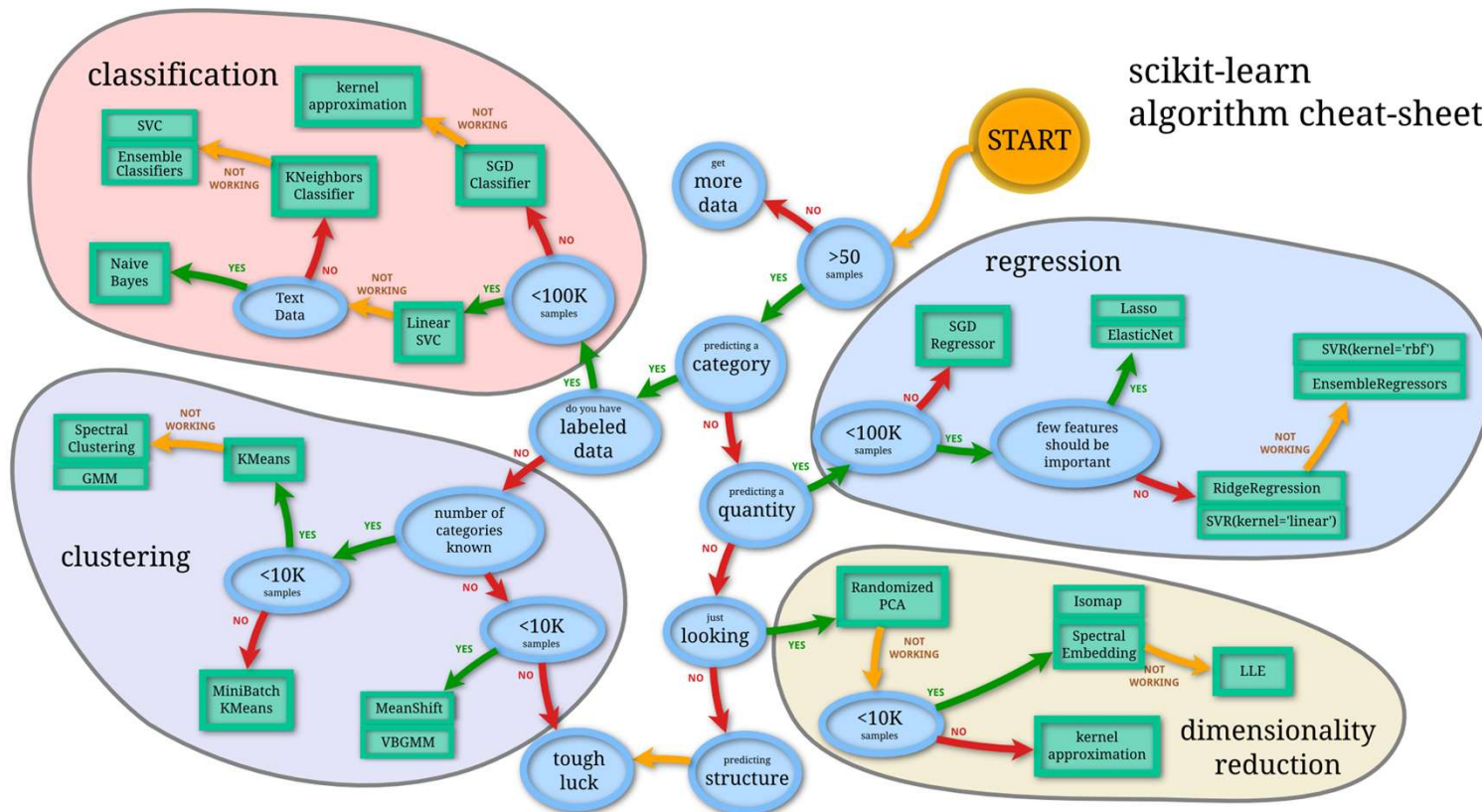
Landscape of Algorithms



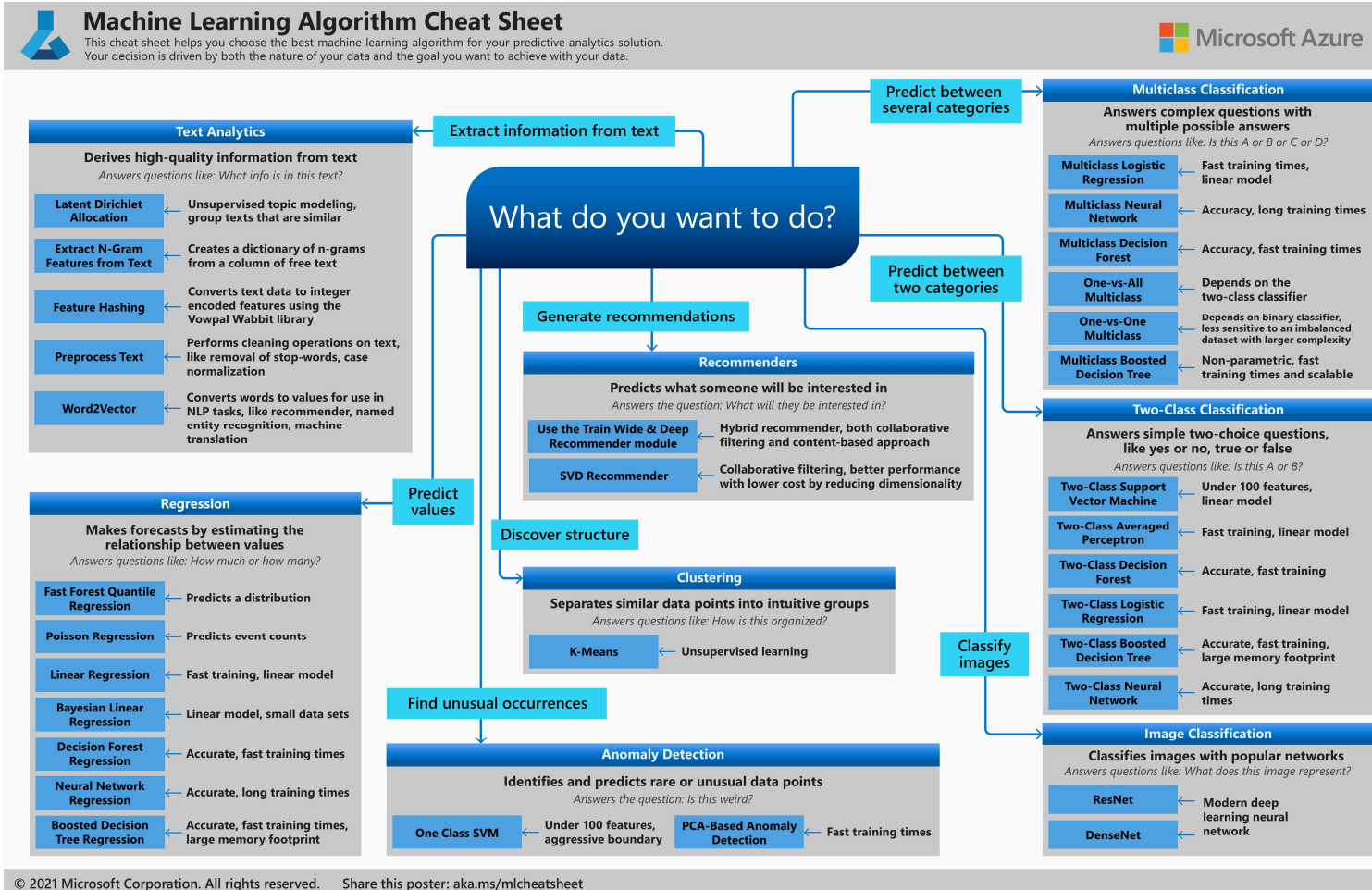
Types of Machine Learning



Choosing the Algorithm



Another take...

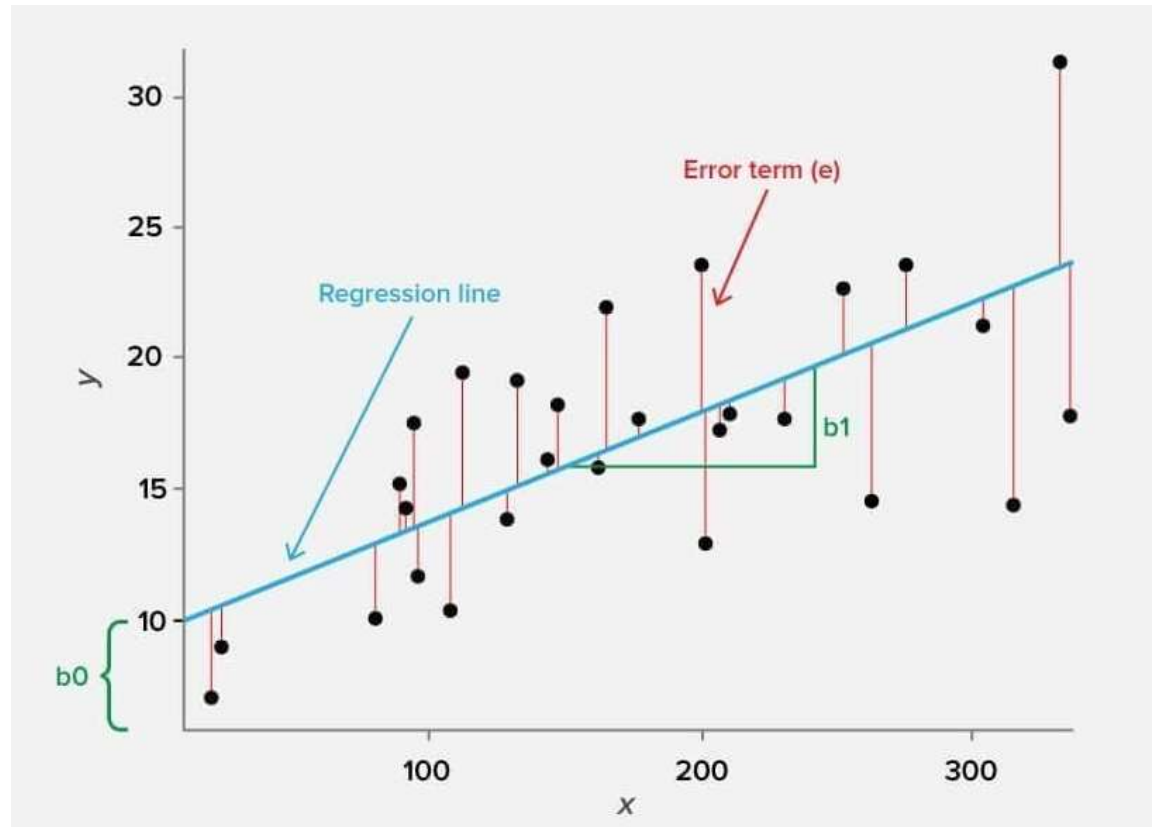


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A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed top and a visible clock face stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground is filled with lush green trees and a wide, paved pedestrian walkway. Several students are walking along the path; some are carrying backpacks, and one student is sitting on a wooden bench on the left. A black lamppost is visible on the left side of the path. The overall atmosphere is that of a busy academic environment.

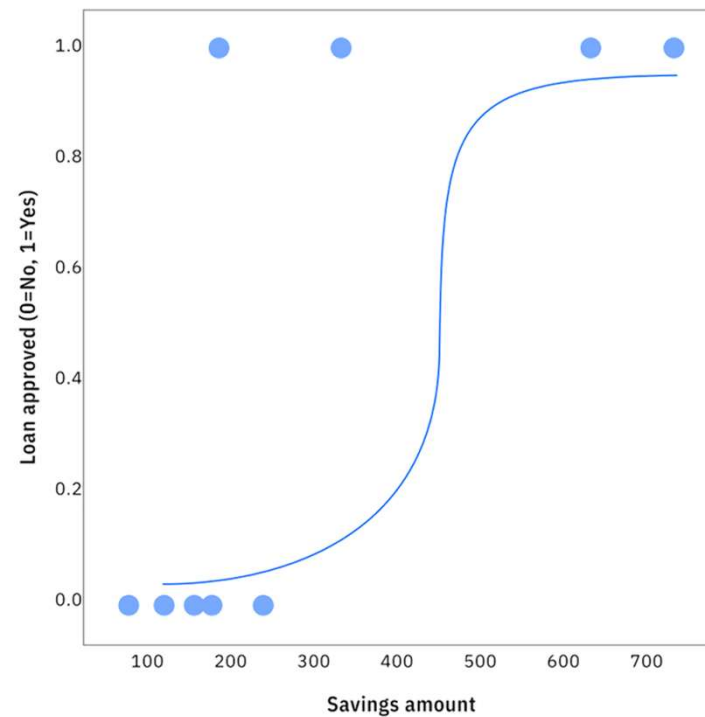
Linear & Logistic Regression

Linear Regression



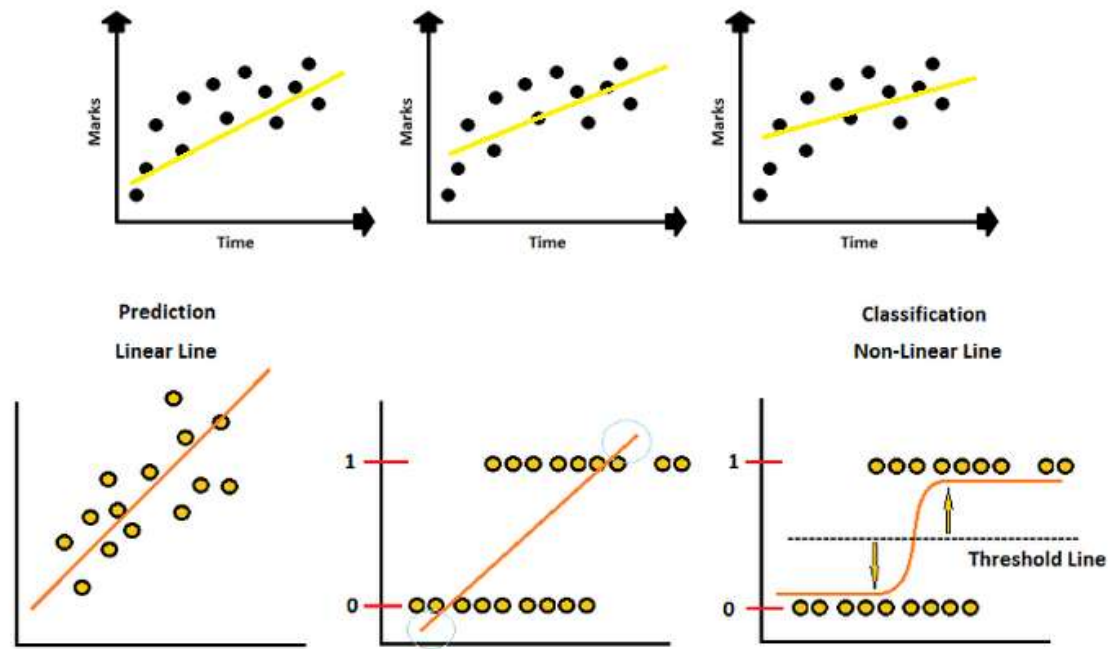
Source: Supervised Learning Algorithms: A Comparison, November 2020 Kristu Jayanti Journal of Computational Sciences (KJCS) - DOI:10.59176/kjcs.v1i1.1259. Free access

Logistic Regression



Source: IBM <https://www.ibm.com/think/topics/logistic-regression>

Linear vs Logistic Regression



Linear vs Logistic Assumptions

Linear Regression Assumptions

1. Linearity
2. No Multicollinearity (predictor correlation)
3. Independence of Observations
4. Normal Distribution of Errors
5. Variance of errors is zero (homoskedasticity)
6. Errors are Independent (no autocorrelation)

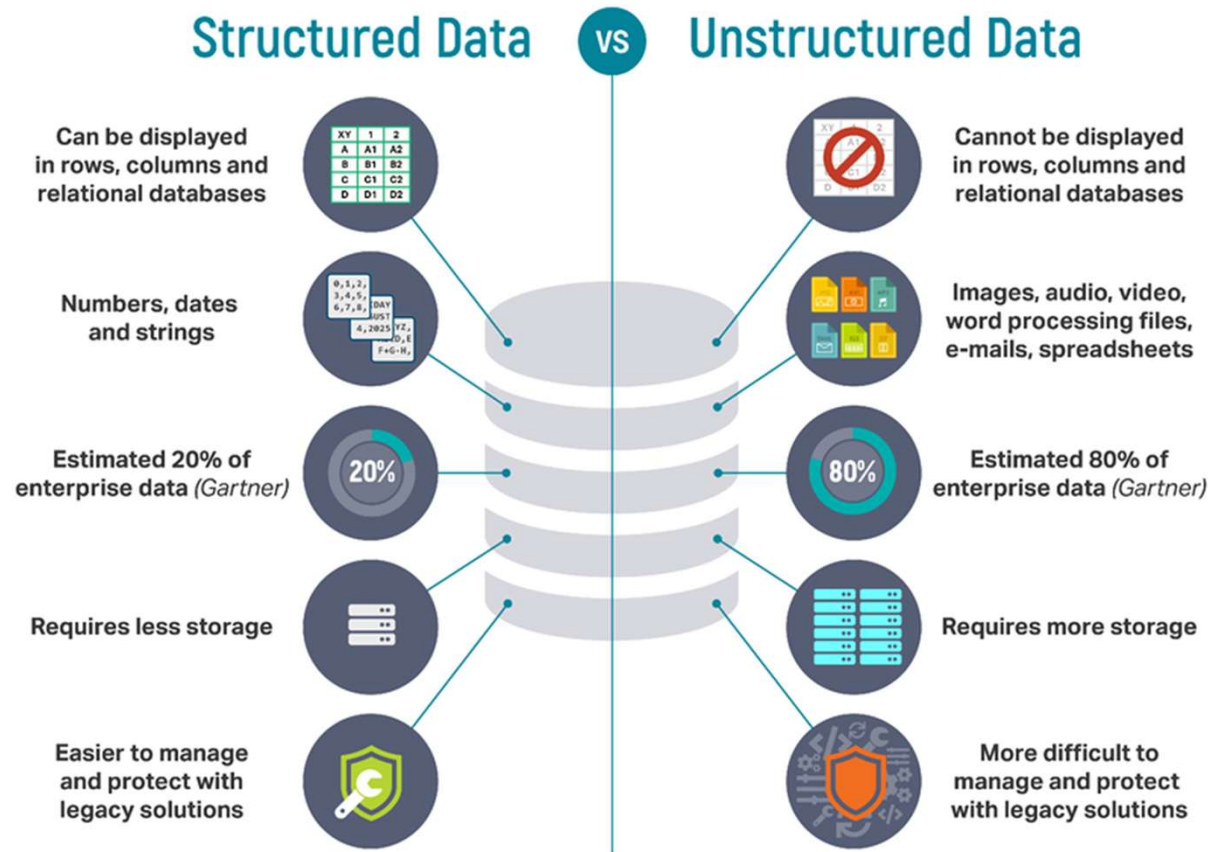
Logistic Regression Assumptions

1. Binary Outcome
2. Log-Linearity
3. Independence of Observations
4. No Multicollinearity
5. Large Sample Size
6. No Outliers

Visualizing Data

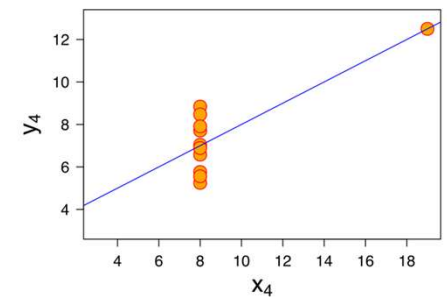
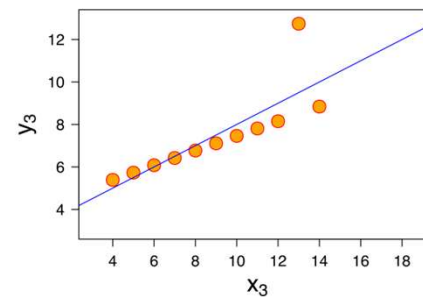
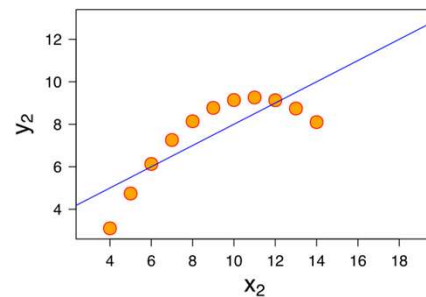
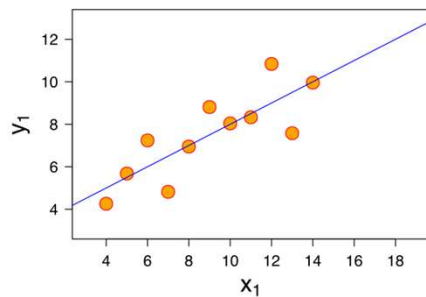


Types of data



Exploratory Data Analysis (EDA)

What do these data sets have in common?



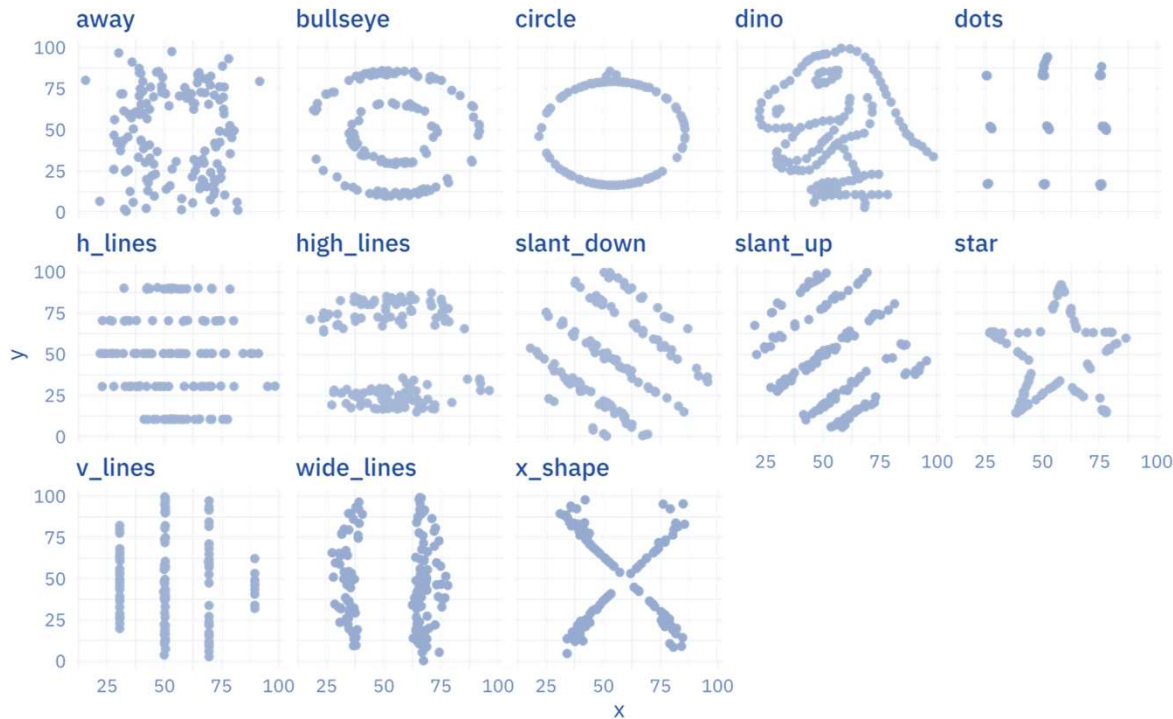
Not much. Except....

Identical means, variances, correlation, coefficients of determination, regressions

Property	Value
Mean of x	9
Sample variance of x: s_x^2	11
Mean of y	7.50
Sample variance of y: s_y^2	4.125
Correlation between x and y	0.816
Linear regression line	$y = 3.00 + 0.500x$
Coefficient of determination of the linear regression: R^2	0.67

Source: Anscombe's quartet from Wikipedia

Another Example

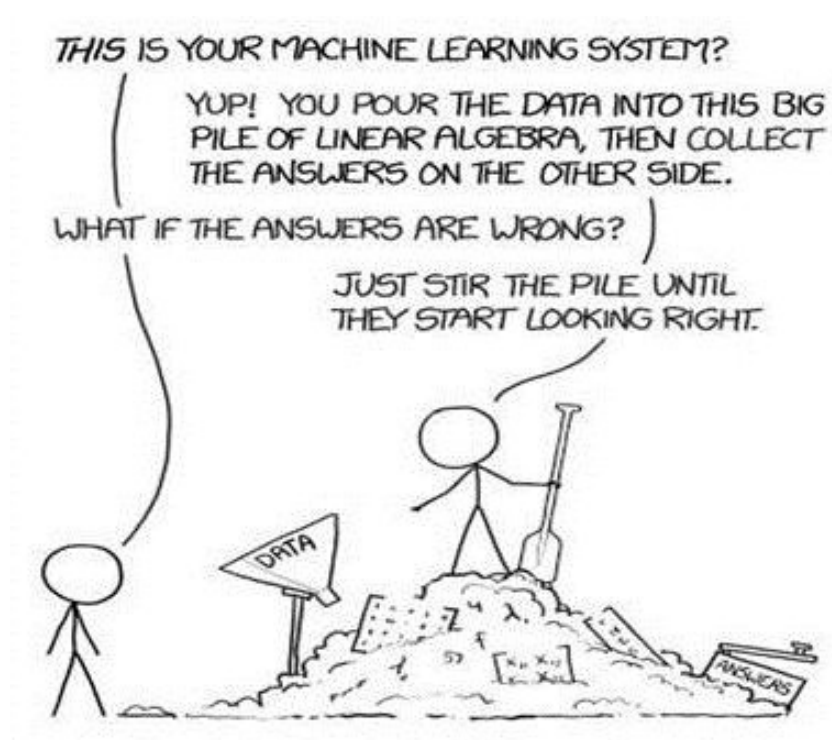


Common statistical values for each group in the dataset							
Dataset	Summary statistics					Regression results	
	Mean x	Mean y	Std Dev x	Std Dev y	Corr x y	Intercept	Coefficients
Away	54.27	47.83	16.77	26.94	-0.06	53.43	-0.10
Bullseye	54.27	47.83	16.77	26.94	-0.07	53.81	-0.11
Circle	54.27	47.84	16.76	26.93	-0.07	53.80	-0.11
Dino	54.26	47.83	16.77	26.94	-0.06	53.45	-0.10
Dots	54.26	47.84	16.77	26.93	-0.06	53.10	-0.10
H_lines	54.26	47.83	16.77	26.94	-0.06	53.21	-0.10
High_lines	54.27	47.84	16.77	26.94	-0.07	53.81	-0.11
Slant_down	54.27	47.84	16.77	26.94	-0.07	53.85	-0.11
Slant_up	54.27	47.83	16.77	26.94	-0.07	53.81	-0.11
Star	54.27	47.84	16.77	26.93	-0.06	53.33	-0.10
V_lines	54.27	47.84	16.77	26.94	-0.07	53.89	-0.11
Wide_lines	54.27	47.83	16.77	26.94	-0.07	53.63	-0.11
X_shape	54.26	47.84	16.77	26.93	-0.07	53.55	-0.11

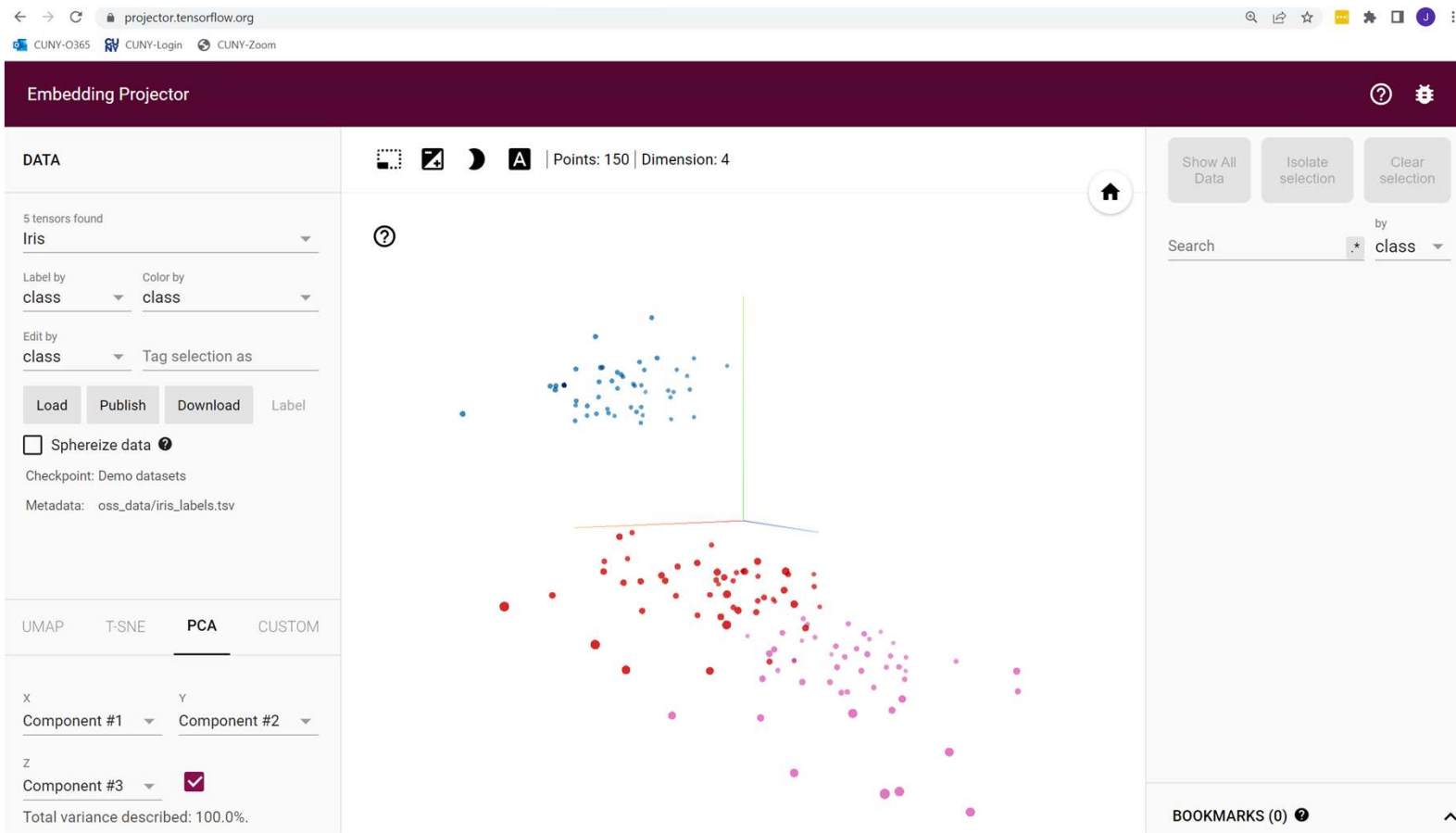
Descriptive statistics can be misleading. Data visualization helps.

Source: Datasaurus Dozen from Wikipedia

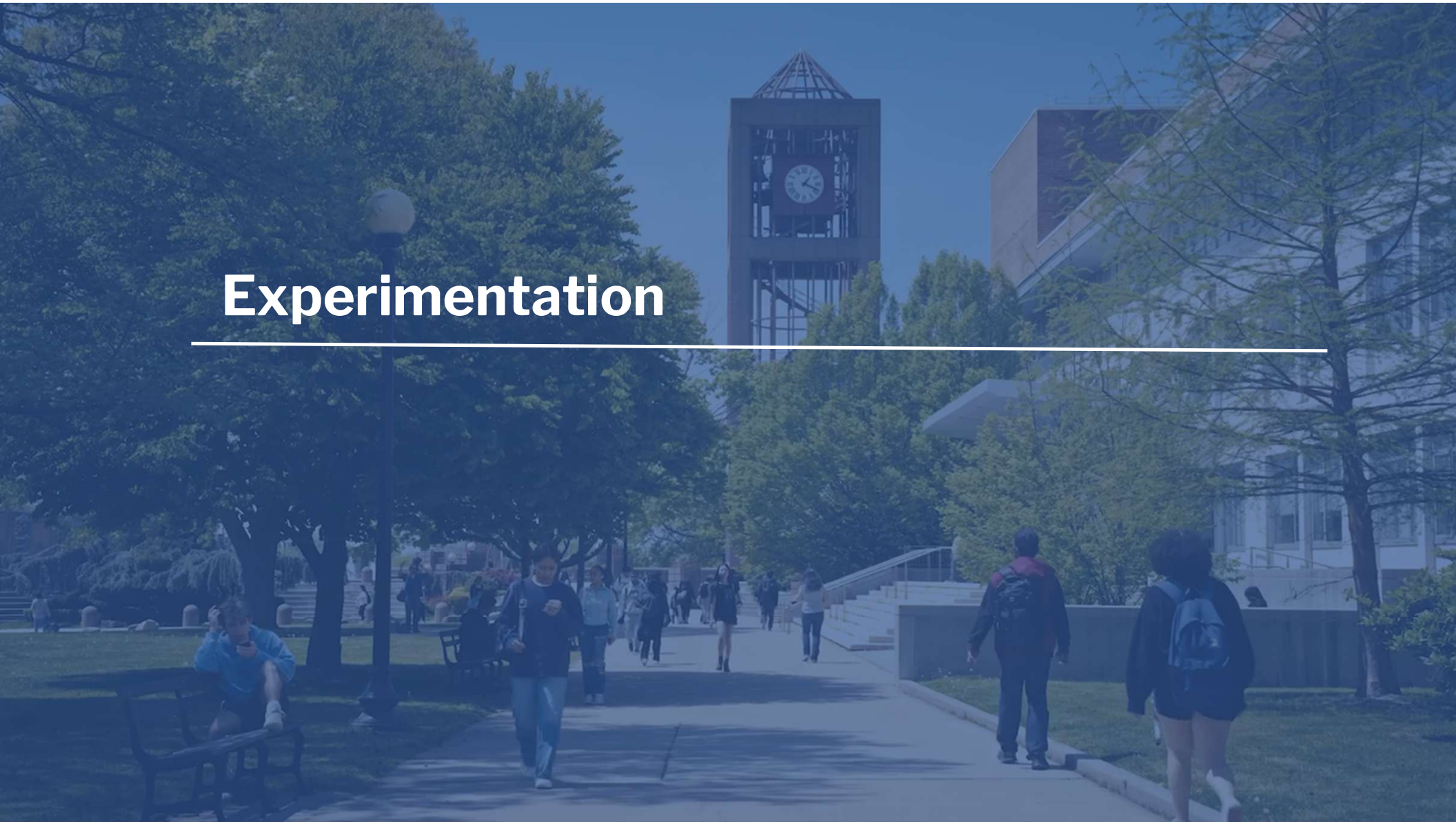
What Data Science isn't...



Visualizing Data

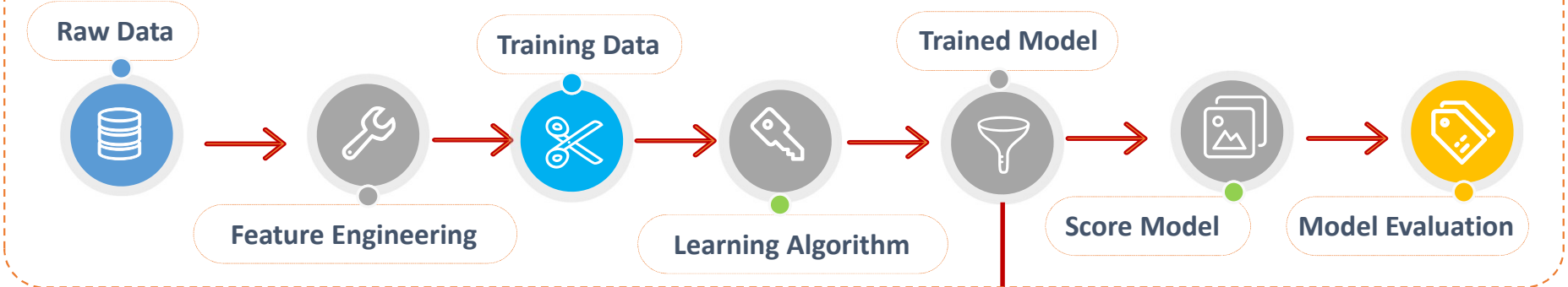


Experimentation

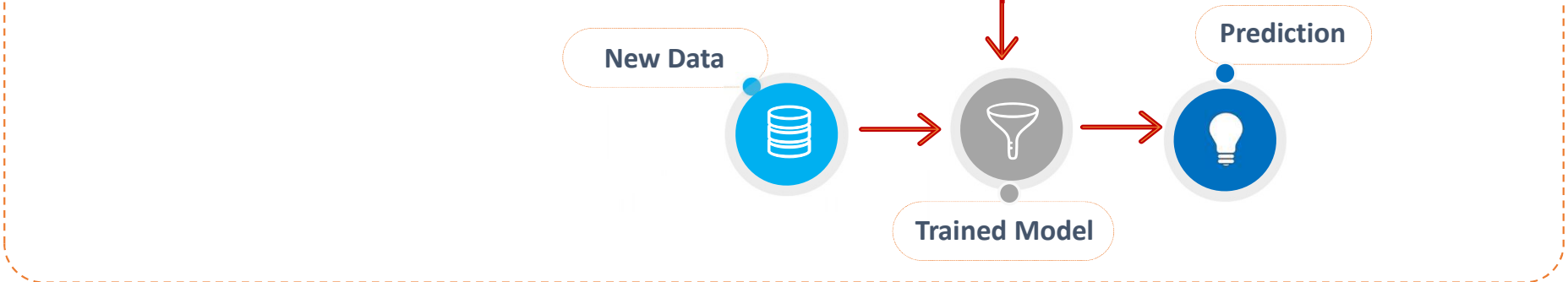


How does Machine Learning work?

TRAINING



INFERENCE

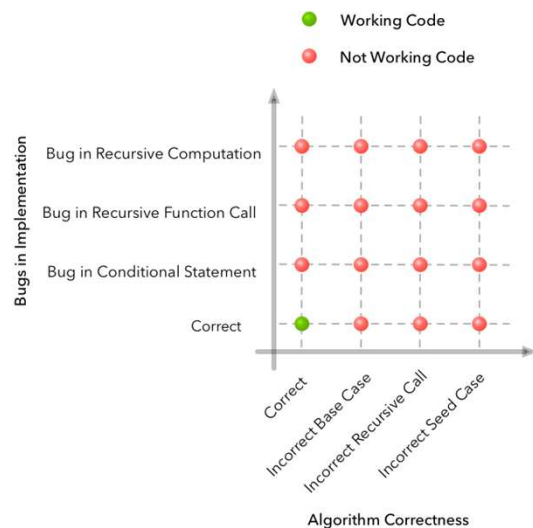


Why is Data Science hard?

Data science is interdisciplinary

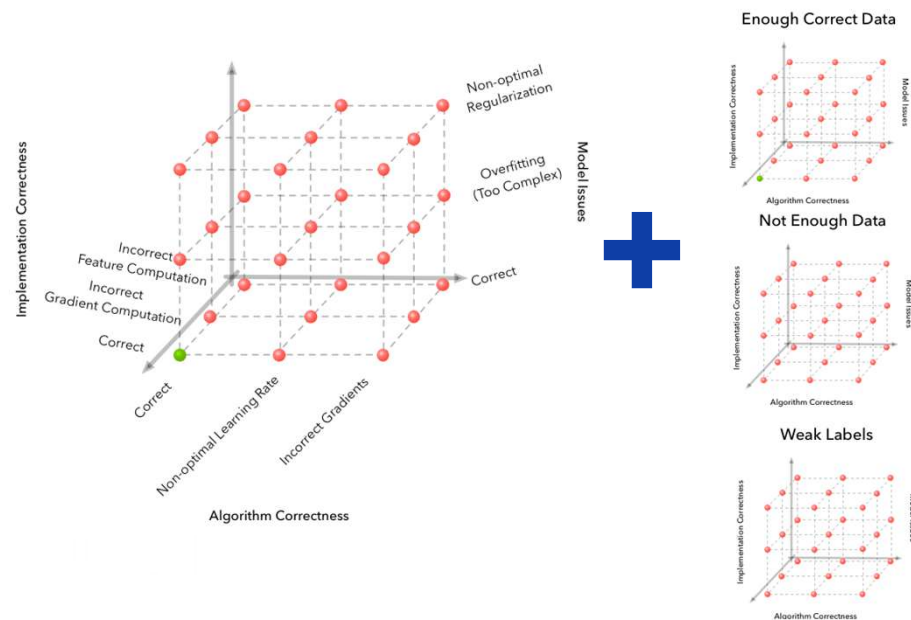
Software Development is a Two-dimensional Problem Space

Algorithm selection and Software implementation



Machine Learning is a Four-dimensional Problem Space

Software development + Model Implementation + Data Signal



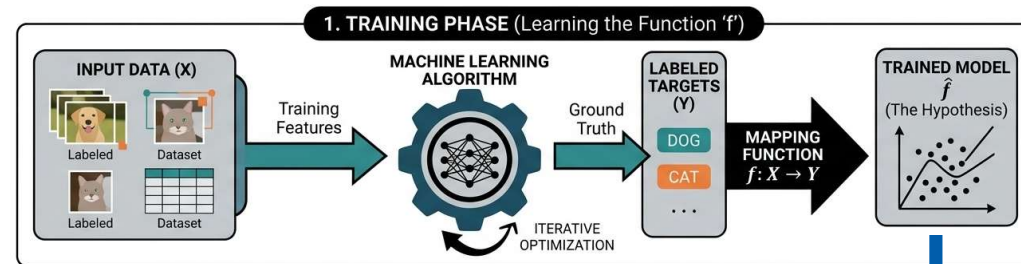
Source: <https://ai.stanford.edu/~zayd/why-is-machine-learning-hard.html>

Why is Data Science complex?

- 1. Data science is interdisciplinary**
- 2. Applicable to many domains (e.g., sciences, finance, healthcare, etc.)**
- 3. Big Data**
- 4. Analysis of data**
- 5. Statistics (a pillar of data science)**
- 6. Experimentation**
- 7. Constant Evolution**
- 8. Data Wrangling**

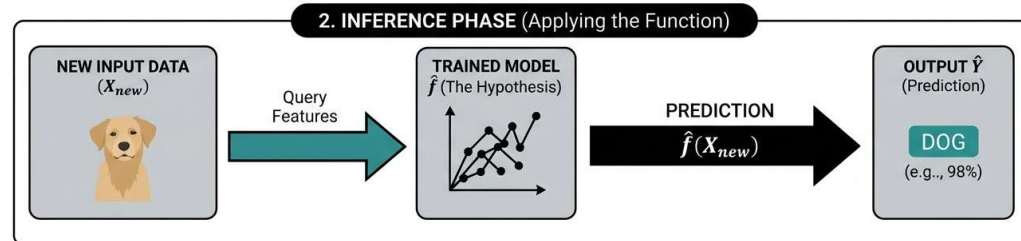
Production Deployment

Training
(Development)

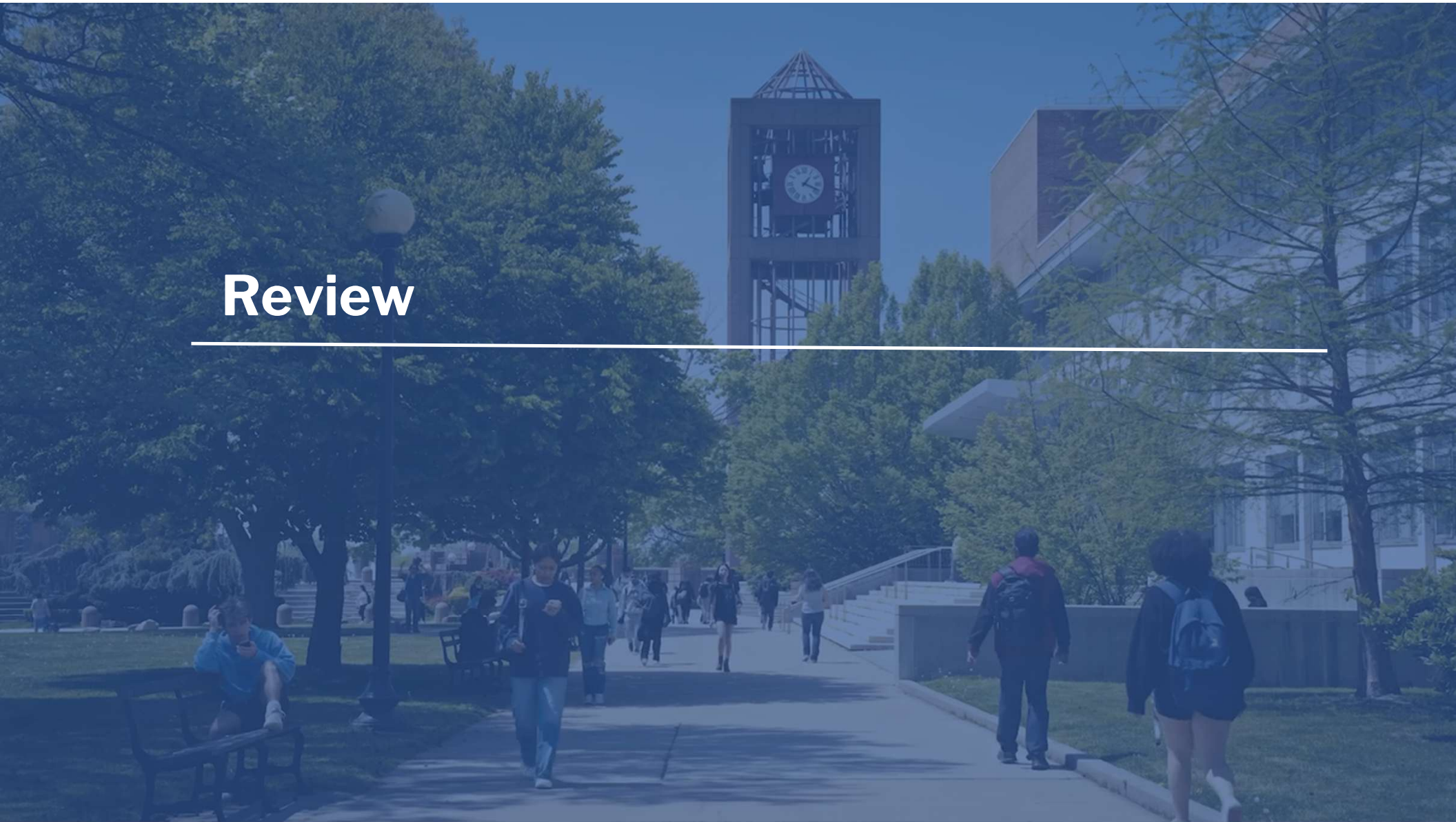


Deployment

Inference
(Production)



Review



This week we covered

Lesson Objectives/Topics

1. Fit and interpret linear regression models to identify relationships between variables
2. Apply logistic regression for binary classification tasks in business scenarios
3. Evaluate model performance using error metrics and classification accuracy
4. Explain the assumptions and limitations of regression techniques

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's steel structure and suspension cables are prominent. The water of the Hudson River is visible in the foreground. The sky is clear and blue.

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